UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II, SESSION 2025/2026

SUBJECT CODE : SECB4243

SUBJECT NAME : SPECIAL TOPIC IN BIOINFORMATICS

SECTION 01

DURATION : 5 DAYS

DATE/DAY : 29 JUNE – 3 JULY 2026

PLATFORM : UTM E-LEARNING

INSTRUCTIONS:

1) This alternative assessment is an **INDIVIDUAL** assessment consisting of **FOUR (4)** questions.
2) Students are required to answer ALL questions.
3) Please attach this page as the front page of your report.
4) Submission to be done through e-learning. Please follow the submission instructions in this booklet.

(Please Write Your Name and Section in Your Answer Booklet)

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WARNING: Disciplinary action will be taken against students who are found cheating during the examination.

This question paper consists of **THIRTEEN (13)** printed pages including this page.

Question 1 - Computational Bioinformatics Concepts & Problem Identification

QUESTION 1 A

I - Background problem

Lung Adenocarcinoma or (LUAD) is a type of non-small lung cancer which is the most common type in lung cancer where it will move more slowly than small cell type. Often found in outer parts of the lung where it can occur to smokers and also non-smokers due to exposure or genetic mutations. Patients that carry similar symptoms might actually have different diseases from each other.

This is because LUAD is dependent on the genetics of each. For example, some people might have mutations in their genes like KRAS, TP53, STK11, or EGFR where these mutations will affect how the genes are expressed and how the cancer is going to behave, whether the treatment is effective or not. Because of that, only looking at one type of data is not enough. Combination of multiple data types such as Gene mutation data, Gene expression data and Clinical data will help seeing a bigger picture where a single data type might have missed.

Many studies focus on one layer only such as focusing on one omics layer only. This makes the studies might have missed out some important details such as its connection with other layers thus why multi omics layers exist in studies but of course when combining multiple data types will make the studies harder.

Ii - Importance of Computational Approaches

The studies in LUAD are very large and complex as it can be time consuming and almost impossible to do this manually. For example, Gene Expression data contain tens of thousands of genes per patient, then try to combine it with mutation data that varies between each patient and combine it with Clinical data such as survival time, stage and patient characteristic.

Computational methods such as Deep Learning helps in making the studies possible where it is able to help in finding genes that behave differently between each stage of cancer, link gene mutation to changes in gene activity and patient outcome and many more.

Iii - Role of Bioinformatics

A bioinformatician's job is to make sure that it acts as a bridge between biological data and meaningful medical insights where their job can be varied from cleaning and organizing biological data so it can be used together, explore patterns in data, build predictive systems and many more.

QUESTION 1 B

The dataset used for this study is TCGA Lung Adenocarcinoma (LUAD) – PanCancer Atlas dataset from cBioPortal. Although it is treated as one dataset, it contains multiple types of data collected from the same group of patients. These include:

- Clinical data (patient information like age, sex, and survival)
- Mutation data (gene-level DNA changes)
- Gene expression data (RNA activity levels)

All of these data types are linked with the same patient or sample ID which makes it possible to integrate it together into one dataset. This makes the dataset suited for this study which is the main goal to combine multiple data types to study how genetic changes affect gene activity and clinical outcome within the same patient. This is important to learn the relation between lung cancer develops and progresses.

Dataset Details

Item	Description
Source	TCGA LUAD PanCancer Atlas (cBioPortal / GDC)
Sample size	Around 500+ patients (final number confirmed after merging datasets)
Clinical data	Age, sex, smoking history, cancer stage, survival status
Mutation data	Gene mutations such as KRAS, TP53, EGFR, STK11
Expression data	Gene activity levels for thousands of genes
Format	CSV / TSV / MAF files linked by patient/sample ID

Biological Relevance and Analytical Use

This dataset is suited for this experiment as it contain different type of information for each of patient. Because of that, we are able to see the relation between genetic mutation with gene activity and patient outcome.For example, we might be able to see mutations like KRAS may be linked to changes in gene expression. This makes it useful for discovering biomarkers and understanding cancer behavior.

Sources

- cBioPortal for Cancer Genomics — TCGA-LUAD (PanCancer Atlas 2018) study, cbioportal.org
- Source Link: https://www.cbioportal.org/study/summary?id=luad_tcga_pan_can_atlas_2018